

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II EXAMINATION – WINTER 2024****Subject Code:BE01000101****Date:18-01-2025****Subject Name:Basic Civil Engineering****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Discuss scope of civil engineering.	03
	(b) Discuss various grades of concrete.	04
	(c) Discuss classification of buildings based on structure.	07
Q.2	(a) Convert the following RB in to W.C.B: 1) N55°30'W 2) S29°00'W 3) S23°00'E	03
	(b) Differentiate between plane surveying and geodetic surveying.	04
	(c) What is FRP? Write uses and characteristics of FRP.	07
	OR	
	(c) Discuss classification of bricks and what are the properties of a good brick?	07
Q.3	(a) Discuss aims and objectives of town planning.	03
	(b) Differentiate between English bond and Flemish bond.	04
	(c) What are the elementary principles of building planning? Describe them briefly.	07
	OR	
Q.3	(a) How would you select the materials for low-cost housing?	03
	(b) Explain site plan and key plan.	04
	(c) Describe the various principles of zoning.	07
Q.4	(a) What are the Indian standards for drinking water?	03
	(b) What are the effects of solid waste?	04
	(c) Discuss various pipes used in the house drainage plumbing system.	07
	OR	
Q.4	(a) Write advantages and disadvantages of BRTS.	03
	(b) Draw a typical plan for drainage of a small house.	04
	(c) Describe pillars of smart city.	07
Q.5	(a) Write applications of surveying.	03
	(b) What do you understand by indirect ranging? Describe in detail with neat sketch.	04
	(c) The following are bearings of a closed traverse ABCDA. Find out interior angles and correct them if necessary.	07

LINE	FB	B.B
AB	55°30'	235°00'
BC	113°00'	293°30'
CD	190°00'	10°00'
DA	279°00'	99°00'

OR

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|------------|------------|---|-----------|
| Q.5 | (a) | Discuss characteristics of contours with suitable sketches. | 03 |
| | (b) | Explain temporary adjustments of level. | 04 |
| | (c) | Describe briefly the application and uses of GPS | 07 |

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I&II EXAMINATION – SUMMER 2025

Subject Code:BE01000101

Date:16-06-2025

Subject Name:Basic Civil Engineering

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Define the civil engineering and enlist the branches of civil engineering.	03
	(b) Convert (i) 10 m ² to ___ ft ² (ii) 5 km to ___ meter (iii) 1 Acre to ___ m ² (iv) 2 cubic meter to _____cu. Ft.	04
	(c) Enlist the various types of cement and discuss the properties with its uses.	07
Q.2	(a) Enlist commonly used smart materials and explain any one.	03
	(b) Enlist principles of planning and describe any one in detail.	04
	(c) Draw cross section through window for a single wall for a residential building from foundation bottom to parapet top showing spread footing foundation for 30 cm thick wall. Label the components and provide dimensions.	07
	OR	
	(c) Draw the symbol of concrete, brick, wood, revolving door And Write the abbreviation of PL, FSI, RCC	07
Q.3	(a) Classify the buildings based on the structure.	03
	(b) Differentiate between BOD and COD.	04
	(c) Enlist various building services and objectives of building services.	07
	OR	
Q.3	(a) Write drinking water quality standards as per BIS.	03
	(b) List the types of foundations. Explain various shallow foundations with sketch.	04
	(c) Define solid waste management. Give objectives and activities of solid waste management.	07
Q.4	(a) Give difference between Surveyors and prismatic compass.	03
	(b) Draw the conventional symbols for Survey station, Temple, Lake, and Double railway line	04
	(c) Determine the value of included angles in the closed traverse ABCD in clockwise direction for the given below bearings. Apply necessary check also.	07

LINE	Fore Bearing
AB	40°
BC	70°
CD	210°
DA	280°

OR

- Q.4** (a) Differentiate between W.C.B and Q.B system. **03**
(b) Enlist the various instruments used in chain survey with sketch. **04**
(c) The following consecutive readings were taken with a level and a 4meter levelling staff on a continuously sloping ground: 0.570, 1.235, 1.750, 2.220, 2.665, 3.410, 1.005, 1.835, 2.165, 3.550, 0.825, 0.965, 1.730 and 2.320. The reduced level of the first point was 100 mt. Calculate reduced level of all stations using the rise and fall method and apply check. **07**
- Q.5** (a) What is Total station? State its field applications. **03**
(b) Describe various uses of GPS technology. **04**
(c) Write short note on BRTS and Metro. **07**
- OR**
- Q.5** (a) Describe important features of green building. **03**
(b) Write a short note on rain water harvesting systems. **04**
(c) Give objectives of riverfront development project and Sardar sarovar project. **07**

Enrollment No./Seat No.:

GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering - SEMESTER - 1/2 EXAMINATION - WINTER 2025

Subject Code: BE01000101/BE01R00101

Date: 30-12-2025

Subject Name: Basic Civil Engineering

Time: 02:30 PM TO 05:00 PM

Total Marks: 70

Instructions

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) List different types of building materials used in construction and state their applications.	03
(b) Explain the properties of cement and aggregates as building materials.	04
(c) Discuss classification of buildings based on structure.	07
Q.2 (a) Differentiate between English bond and Flemish bond.	03
(b) Enlist and explain any two principles of planning of buildings.	04
(c) Explain the concepts of low-cost housing and building automation with examples.	07
OR	
(c) Describe the various principles of zoning.	07
Q.3 (a) List the quality characteristics of drinking water.	03
(b) Differentiate between domestic wastewater and drinking water standards.	04
(c) Explain the methods of wastewater disposal and give a brief note on solid waste management.	07
OR	
(a) Differentiate between W.C.B and Q.B system.	03
(b) What is FRP? Write its uses.	04
(c) Explain different water conservation techniques applicable to urban areas.	07
Q.4 (a) Write applications of surveying.	03
(b) Explain the instruments used for linear measurements with neat sketches.	04

- (c) The following are bearings of a closed traverse ABCDA. Calculate interior angles and correct them if necessary.

07

LINE	F.B	B.B
AB	45°00'	225°00'
BC	123°30'	303°30'
CD	181°00'	1°00'
DA	289°00'	109°00'

OR

- (a) Write the applications of Contour map.

03

- (b) Write short notes on: Local attraction

04

- (c) The following sequential readings were taken on a sloping ground. The first reading was taken on a B.M. of 50.00 m. Calculate R.L. of each point and apply arithmetical check. 0.450, 0.560, 0.785, 1.200, 2.500, 3.460, 0.320, 0.656, 0.985, 1.300.

07

- Q.5 (a) List the features of a smart city.

03

- (b) Explain the applications of drone survey.

04

- (c) What are the three segments of GPS? Describe them briefly.

07

OR

- (a) Discuss aims and objectives of town planning.

03

- (b) Explain the concept of green building with its advantages.

04

- (c) Write a case study on water conservation projects such as Sardar Sarovar Dam, Kalpsar Project, or Riverfront Development.

07
